A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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Abstract

Large language model (LLM) calibration is nearly always reported as a single aggregate scalar, obscuring whether miscalibration is diffuse or concentrated in specific knowledge domains. We develop a statistically rigorous audit framework and apply it to the MMLU benchmark [Hendrycks et al., 2021] using token-level log probabilities from GPT-4o-mini and Llama-3-8B-Instruct. Our framework combines a three-level mixed effects model to decompose calibration variance across domains, subjects, and questions; isotonic and kernel-smoothed reliability curves per subject; empirical Bayes shrinkage of subject-level estimates; and Benjamini–Hochberg FDR correction to identify robustly miscalibrated subjects. For GPT-4o-mini, we find that 7.9% of total calibration gap variance is attributable to between-subject differences, that all 57 subjects are significantly overconfident after FDR correction (mean gap = 0.196), and that question-level entropy and word count are significant positive predictors of miscalibration magnitude. Subject-level gaps range from 0.037 (marketing) to 0.487 (moral scenarios), demonstrating substantial heterogeneity masked by aggregate ECE. For Llama-3-8B-Instruct, the same universal overconfidence is observed (mean gap = 0.197, 57/57 subjects rejected at FDR $\alpha = 0.05$) with 2.0% between-subject ICC, and no question-level features reach significance. Across models, subject-level ECE rankings are highly correlated (Spearman $\rho = 0.846$, 95% CI $[0.734, 0.910]$), indicating that miscalibration is task-driven rather than model-specific.

1 Introduction

Large language models are increasingly deployed in high-stakes settings—healthcare, law, education—where the reliability of model-expressed confidence is as important as accuracy itself. A well-calibrated model is one whose stated confidence tracks its empirical accuracy: if a model assigns 80% probability to an answer, it should be correct 80% of the time. Miscalibration, particularly overconfidence, means that users who rely on model confidence as a decision signal will be systematically misled.

A substantial literature has studied LLM calibration. Guo et al. [2017] established the modern empirical study of neural network calibration, popularizing expected calibration error (ECE) as a standard metric and demonstrating the effectiveness of temperature scaling as a post-hoc correction. Kadavath et al. [2022] showed that for multiple-choice tasks, token-level log probabilities provide a meaningful calibration signal, and that base (pre-RLHF) models exhibit better-calibrated token probabilities than instruction-tuned counterparts. Xiong et al. [2024] provided a systematic benchmark of confidence elicitation methods, finding that all methods "struggle in challenging tasks, such as those requiring professional knowledge"—suggesting calibration quality may vary by domain, though the authors do not investigate this directly. Jiang et al. [2021] similarly found that question-level features such as answer entropy predict calibration quality, a pattern we examine formally here.

What is missing is a *statistical audit* of that heterogeneity. To our knowledge, the existing literature reports subject-level ECE without (i) stabilizing estimates against large differences in per-subject sample size, (ii) formally testing which subjects are robustly miscalibrated after correcting for multiple comparisons, (iii) decomposing how much total calibration variance is attributable to domain versus subject versus individual question, or (iv) characterizing subjects by their *pattern* of miscalibration rather than its scalar magnitude.

This framing is analogous to the gap between per-group accuracy reporting and a rigorous fairness audit [Barocas et al., 2023]: the fairness literature long ago recognized that single-number summaries are insufficient for subgroup auditing. This paper imports that logic into calibration evaluation. The answer has downstream consequences: if miscalibration is diffuse, a single global recalibration (e.g., temperature scaling) is a reasonable fix. If it is concentrated in specific domains—professional law, abstract mathematics, clinical medicine—then users in those domains face systematically higher risk, and domain-aware recalibration would be warranted.

2 Related Work

Calibration of neural networks. Guo et al. [2017] showed that modern neural networks are systematically overconfident and that temperature scaling largely corrects this at the aggregate level, with Zadrozny and Elkan [2002] earlier establishing isotonic regression as an alternative nonparametric recalibration method. Kadavath et al. [2022] extended calibration evaluation to LLMs, establishing that token-level probabilities on multiple-choice tasks carry a meaningful calibration signal. Nakkiran et al. [2025] and Yaldiz et al. [2026] study how instruction tuning affects calibration, finding that RLHF-style fine-tuning tends to degrade calibration relative to base models, an effect also documented in Ouyang et al. [2022].

Domain-level heterogeneity. Xiong et al. [2024] is the closest prior work in spirit, benchmarking confidence elicitation across task types. Jiang et al. [2021] showed that question-level features predict calibration on QA tasks. Luo et al. [2025] fits subject-specific temperature parameters for each of MMLU's 57 subjects, implicitly acknowledging that a global correction is insufficient; Tan et al. [2026] uses base model signals for unsupervised recalibration. Neither work performs a formal statistical audit of subject-level heterogeneity.

Multiple testing and empirical Bayes. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] controls the false discovery rate and is standard in high-dimensional inference. Empirical Bayes shrinkage of group-level estimates [e.g., Efron, 2012] is standard in multilevel modeling via best linear unbiased predictors (BLUPs). We use the multilevel modeling framework of Raudenbush and Bryk [2002] to decompose calibration variance across levels of the MMLU hierarchy.

3 Data

MMLU. We use the Massive Multitask Language Understanding benchmark [Hendrycks et al., 2021], accessed via the `cais/mmlu` dataset on HuggingFace. MMLU comprises 14,042 four-option multiple-choice questions across 57 subjects nested in four broad domains: STEM, Social Sciences, Humanities, and Other. Subject sizes range from 100 to 1,534 questions (median 163). The three-level hierarchy—questions $\rightarrow$ subjects $\rightarrow$ domains—is natural for the proposed analysis. We use the test split throughout.

Models. We query two instruction-tuned models to enable a comparable cross-model analysis. **GPT-4o-mini** is queried via the OpenAI Batch API at temperature 0 (`max_tokens=1`, `top_logprobs=20`). Using 20 top logprobs—rather than the default 5—ensures that all four answer tokens (A, B, C, D) are captured in the API response for virtually every question; with only 5 logprobs, we found that 55% of questions had at least one answer token missing, artificially inflating renormalized confidence. **Llama-3-8B-Instruct** is run locally using 4-bit quantization (via BitsAndBytes; Dettmers et al. 2022) on a GPU. Because instruction-tuned models tend to begin responses with explanatory text ("The answer is...") rather than a bare letter, we pre-fill the assistant turn with "The answer is" before extracting logprobs, placing the scoring position immediately before the answer token. Both models use the same system prompt instructing a single-letter response.

4 Methods

4.1 Confidence Elicitation

For each question, we extract the token-level log probabilities over the four answer tokens (A, B, C, D). Applying softmax renormalization yields a proper probability distribution over options; the model's confidence $c_i \in (0, 1]$ is the maximum of this renormalized distribution, equal to $P(\text{predicted answer} \mid \text{answer} \in \{A, B, C, D\})$. This is the standard approach in calibration research [Kadavath et al., 2022] and avoids the noise of verbal elicitation.

For GPT-4o-mini, logprobs are drawn directly from the API's top-20 response. For Llama-3-8B-Instruct, logprobs are extracted from the full vocabulary softmax at the assistant-generation position after the "The answer is" prefix, ensuring the model is scored at the position where it selects a letter. Both pipelines compute the same quantity: $P(\hat{y} \mid \hat{y} \in \{A, B, C, D\})$.

4.2 Outcome and Covariates

Primary outcome. The primary outcome is the *signed calibration gap* at the question level:

$$\text{gap}_i = c_i - y_i, \tag{1}$$

where $y_i \in \{0, 1\}$ indicates correctness. This quantity is positive under overconfidence and negative under underconfidence, and supports regression-based inference that binning-based ECE cannot. Its mean at the subject level directly measures the direction and magnitude of miscalibration.

Question-level covariates. We extract the following features from each question:

- **Word count**: number of words in the question stem.
- **Max option length**: word count of the longest answer choice.
- **Negation indicator**: binary, flagging presence of *not*, *except*, *least*, *never*, or *no*.
- **Entropy**: Shannon entropy of the softmax distribution over A/B/C/D, measuring how spread the model's probability mass is across options.

All continuous covariates are standardized before model fitting.

Heteroskedasticity note. The variance of $\text{gap}_i = c_i - y_i$ is not constant across subjects: since $y_i \sim \text{Bernoulli}(p_j)$, its contribution to variance is $p_j(1-p_j)$, which is highest for subjects near 50% accuracy. We check residual plots for evidence of heteroskedasticity and note this as a potential limitation of the homoskedastic mixed model.

4.3 Nonparametric Calibration Curves

For each subject, we estimate a reliability curve using isotonic regression—a nonparametric monotone fit of empirical accuracy on confidence [Zadrozny and Elkan, 2002]—rather than arbitrary equal-width binning. The monotonicity constraint is theoretically motivated but is an assumption rather than a guarantee. We therefore fit an unconstrained kernel smoother (Gaussian, $\sigma = 1.5$ bins) alongside each isotonic curve; substantive divergences between the two identify subjects where the monotonicity assumption may distort the estimated curve.

ECE per subject is computed as the weighted mean absolute deviation between the reliability curve and the diagonal, using 20 equal-width confidence bins.

4.4 Multilevel Regression and Variance Decomposition

We model the question-level calibration gap across all 14,042 questions using a three-level mixed model with questions nested in subjects nested in domains:

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^{\top}\boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \tag{2}$$

where $\mathbf{x}_{ijk}$ are standardized question-level covariates, $u_k \sim \mathcal{N}(0, \sigma^2_{\text{domain}})$ is a domain random intercept, $v_{jk} \sim \mathcal{N}(0, \sigma^2_{\text{subject}})$ is a subject-within-domain random intercept, and $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2_{\varepsilon})$ is residual noise. The model is estimated by restricted maximum likelihood (REML) using `statsmodels.MixedLM` [Raudenbush and Bryk, 2002].

The fixed effects $\boldsymbol{\beta}$ characterize which question features are associated with miscalibration across the full dataset, interpreted as descriptive associations rather than causal effects. The variance components $(\sigma^2_{\text{domain}}, \sigma^2_{\text{subject}}, \sigma^2_{\varepsilon})$ answer RQ1 via an intraclass correlation (ICC) decomposition: the fraction of total calibration heterogeneity attributable to each level of the hierarchy. Subject-level shrunken calibration estimates are the fitted random effects $\hat{v}_{jk}$ (BLUPs), which implicitly borrow strength across subjects in proportion to sample size.

4.5 Multiple Testing Correction

For each of the 57 subjects we test H_0: mean calibration gap $= 0$ (no systematic miscalibration) against the two-sided alternative using a one-sample t-test on the question-level gaps. We apply the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] at $\alpha = 0.05$ to control the false discovery rate, yielding a ranked list of subjects where miscalibration is statistically established rather than just descriptively apparent.

4.6 Cross-Model Replication

The full pipeline is run on both GPT-4o-mini and Llama-3-8B-Instruct. Subjects flagged as miscalibrated by both models suggest task-driven miscalibration reflecting question structure; subjects that diverge suggest model-specific effects of training. We compare subject-level ECE rankings across models using Spearman rank correlation with bootstrap confidence intervals (2000 replicates, seed 42).

5 Results

5.1 Variance Decomposition (RQ1)

Table 1 reports the ICC decomposition from the three-level mixed model for both models. For GPT-4o-mini, the large majority of calibration gap variance (92.2%) is at the question level, consistent with high within-subject variability in model confidence. Between 7.9% of variance is attributable to systematic differences across subjects, indicating that some subjects are persistently more miscalibrated than others—not merely unlucky draws from a common distribution. For Llama-3-8B-Instruct, the between-subject ICC is substantially smaller at 2.0%, suggesting that Llama's miscalibration is more diffuse across subjects. Domain-level variance collapses to zero for both models.

Table 1: Variance decomposition of the calibration gap for both models.

	GPT-4o-mini		Llama-3-8B-Instruct	
Level	Variance	ICC	Variance	ICC
Domain	0.000	0.000	0.000	0.000
Subject	0.013	0.079	0.004	0.020
Residual	0.150	0.922	0.190	0.980

The domain-level variance is estimated at zero for both models. With only four domains, the likelihood is flat with respect to the domain variance component, and the estimate collapses to the boundary. We therefore report total between-subject variance without a clean domain/subject partition, and treat domain as a descriptive grouping rather than a modeled random effect.

5.2 Calibration Curves by Subject

Figure 1 shows reliability curves (isotonic and kernel) for the five most miscalibrated subjects by ECE for GPT-4o-mini. All five exhibit the characteristic shape of an overconfident model: the reliability curve lies below the diagonal, indicating that the model's stated confidence systematically exceeds its empirical accuracy. The isotonic and kernel curves are in close agreement for most subjects, validating the monotonicity assumption. Subjects in the humanities domain (moral scenarios, professional law, formal logic) show the largest absolute gaps.

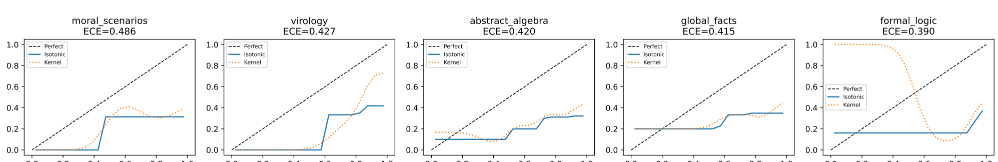

Figure 1: Reliability curves for the five most miscalibrated GPT-4o-mini subjects by ECE. Isotonic (solid) and kernel-smoothed (dashed) estimates are shown alongside the calibration diagonal. All five subjects are strongly overconfident.

5.3 Predictors of Miscalibration (RQ2)

Table 2 reports the fixed effects from the multilevel model. The intercept of 0.196 ($p < 0.001$) confirms strong, uniform overconfidence across all subjects for GPT-4o-mini, consistent with the documented effect of RLHF fine-tuning on calibration [Ouyang et al., 2022, Nakkiran et al., 2025].

Among question-level predictors, **word count** ($\hat{\beta} = 0.031$, $p < 0.001$) and **entropy** ($\hat{\beta} = 0.016$, $p < 0.001$) are both significantly positive. Longer questions are associated with larger calibration gaps, plausibly because complexity increases overconfidence. Higher entropy—indicating the model spreads probability more evenly across options—is also associated with larger gaps, suggesting that ambiguous questions exacerbate miscalibration. Neither max option length nor the negation indicator reaches significance.

Table 2: Fixed effects from the three-level mixed model for both models. All continuous covariates are standardized. Significance: $^{***}p < 0.001$, $^{\dagger}p < 0.10$.

	GPT-4o-mini				Llama-3-8B-Instruct			
Predictor	$\hat{\beta}$	SE	z	p	$\hat{\beta}$	SE	z	p
Intercept	0.196	0.016	12.55	<0.001***	0.197	0.010	20.58	<0.001***
Word count	0.031	0.006	5.20	<0.001***	-0.005	0.006	-0.78	0.437
Max option length	0.001	0.004	0.19	0.848	-0.001	0.005	-0.20	0.838
Negation	-0.001	0.010	-0.06	0.950	0.007	0.011	0.63	0.530
Entropy	0.016	0.004	4.30	<0.001***	0.008	0.004	1.82	$0.069^{\dagger}$

Figure 2 shows the coefficient forest plot for both models. The contrast between models is striking: while GPT-4o-mini shows significant positive effects of word count and entropy, none of Llama's predictors reach significance at $\alpha = 0.05$. Entropy is marginally positive ($p = 0.069$) in the same direction as GPT, suggesting a weak shared signal, but the effect is much attenuated. This difference may reflect Llama's lower ICC (2.0% vs. 7.9%): with less between-subject structure, there is less systematic variance for predictors to explain.

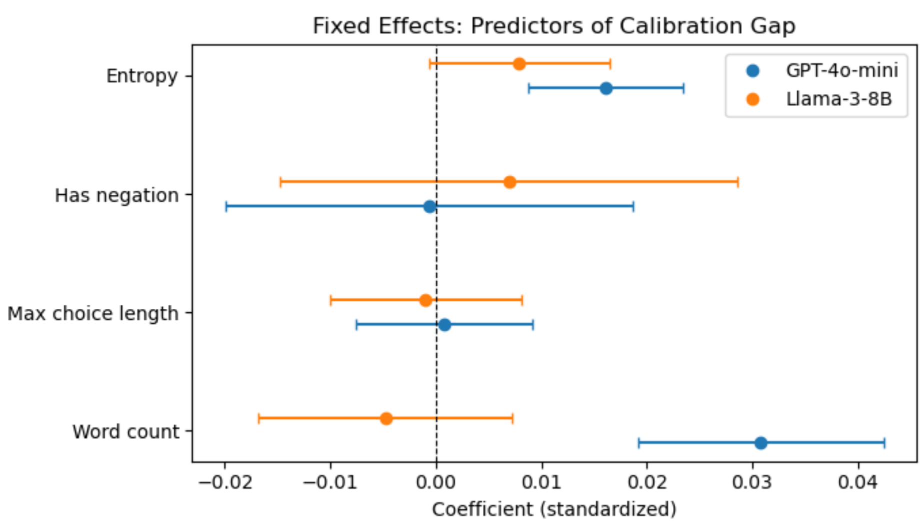

Figure 2: Fixed effect estimates with 95% confidence intervals from the multilevel model. Estimates are on the standardized calibration gap scale.

5.4 FDR-Corrected Subject Rankings

Figure 3 shows all 57 subjects sorted by mean calibration gap, with FDR-rejected subjects (all 57 for both models) highlighted. All 57 subjects are rejected at FDR $\alpha = 0.05$ for both GPT-4o-mini and Llama-3-8B-Instruct. This universal rejection reflects two compounding factors: the intercepts of 0.196 and 0.197 are large relative to the per-subject standard errors, yielding large t-statistics for the average subject; and the BH procedure, which adjusts for 57 tests, still leaves p-values near zero throughout.

The substantive finding is therefore the *magnitude* of heterogeneity, not the binary reject/fail-to-reject outcome. Gaps range from 0.037 (marketing) to 0.487 (moral scenarios)—a 13-fold range. Subjects in the humanities domain (moral scenarios, professional law, formal logic, abstract algebra) cluster at the upper end; social science and STEM subjects show more moderate gaps; and several subjects in the Other category (marketing, medical genetics, college biology) show the smallest gaps. This pattern is consistent with RLHF training data being skewed toward applied, factual content that may be better represented in pretraining.

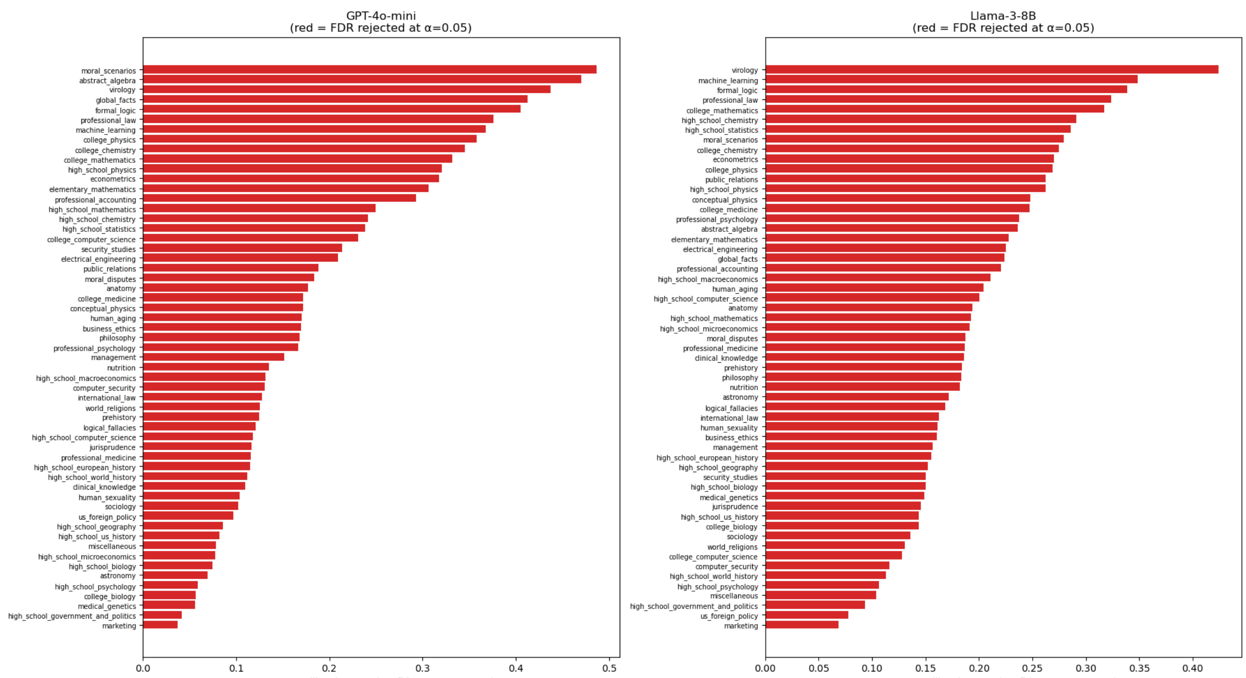

Figure 3: Mean calibration gap per subject, sorted ascending. Red bars indicate subjects where H_0: mean gap $= 0$ is rejected at FDR $\alpha = 0.05$. Both GPT-4o-mini (left) and Llama-3-8B-Instruct (right) show 57/57 subjects rejected, though gap magnitudes and subject ordering differ between models.

5.5 Sensitivity Analysis

We assess the stability of our three main findings under changes to modeling choices.

ECE binning and proper scoring rules. ECE depends on the number of confidence bins used for integration. We recompute subject-level ECE for GPT-4o-mini across bin counts $b \in \{10, 15, 20, 25, 30\}$ and compare each subject ranking to the 20-bin baseline via Spearman rank correlation. All correlations exceed 0.997, and mean ECE varies only from 0.190 ($b = 10$) to 0.194 ($b = 30$). Furthermore, to ensure our subject-level rankings are not artifacts of binning schemes, we evaluate the Negative Log Calibration Score (NLCS)—a strictly proper scoring rule computed as the binary log loss of the model's confidence. The rank correlation between binned ECE and unbinned NLCS remains extremely high (Spearman $\rho \geq 0.903$ across all bin choices). Subject rankings are essentially identical across metrics and bin choices, confirming that the top-5 miscalibrated subjects and their relative ordering are robust.

FDR threshold. The 57/57 rejection finding could in principle be sensitive to the choice of $\alpha = 0.05$. We reapply the Benjamini–Hochberg procedure at $\alpha \in \{0.01, 0.05, 0.10\}$. At the strictest threshold ($\alpha = 0.01$), 56 of 57 subjects are still rejected; at $\alpha = 0.05$ and $\alpha = 0.10$, all 57 are rejected. The near-universal overconfidence finding is not sensitive to the significance threshold.

Mixed model structure. We compare the three-level model (domain + subject random effects) to a two-level model with only subject random effects, to check whether collapsing the domain level changes the estimated between-subject ICC. Both models yield $\hat{\sigma}^2_{\text{subject}} = 0.0128$ and $\text{ICC}_{\text{subject}} = 0.079$. This is expected given that domain-level variance is estimated at zero in the three-level fit—the two specifications are numerically equivalent for GPT-4o-mini. The 7.9% between-subject variance share is therefore robust to the hierarchical structure assumed.

Results for Llama-3-8B-Instruct are consistent. ECE subject rankings are stable across bin counts and correlate strongly with NLCS (Spearman $\rho \geq 0.861$). At $\alpha = 0.01$, 55 of 57 subjects are rejected (the two exceptions have adjusted p-values of 0.014 and 0.031, just above the stricter threshold); at $\alpha = 0.05$ and $\alpha = 0.10$, all 57 are rejected. The two-level and three-level models again yield identical ICC estimates ($\text{ICC}_{\text{subject}} = 0.020$). The main quantitative findings are therefore robust for both models.

5.6 Cross-Model Agreement (RQ3)

Figure 4 shows the per-subject ECE scatter for GPT-4o-mini vs. Llama-3-8B-Instruct, colored by agreement category. The Spearman rank correlation of subject-level ECE is $\rho = 0.846$ (bootstrap 95% CI $[0.734, 0.910]$), indicating strong agreement in which subjects are most miscalibrated across the two very different model families. Because all 57 subjects are flagged as miscalibrated for both models, the correlation of calibration gaps provides the primary evidence of task-driven miscalibration.

The strong cross-model correlation supports the interpretation that subject-level miscalibration is primarily a property of the *task*—the knowledge domain and question structure—rather than of any particular model's training. Both RLHF fine-tuning of a large proprietary model (GPT-4o-mini) and supervised fine-tuning of a smaller open-weights model (Llama-3-8B-Instruct) produce the same pattern of domain-specific overconfidence, with humanities and abstract reasoning subjects consistently most affected across both models.

6 Discussion

Overconfidence is universal and heterogeneous across both models. Both GPT-4o-mini and Llama-3-8B-Instruct show universal overconfidence (57/57 subjects rejected, intercepts of 0.196 and 0.197 respectively), consistent with the documented effect of instruction fine-tuning on calibration [Ouyang et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]. However, the magnitude

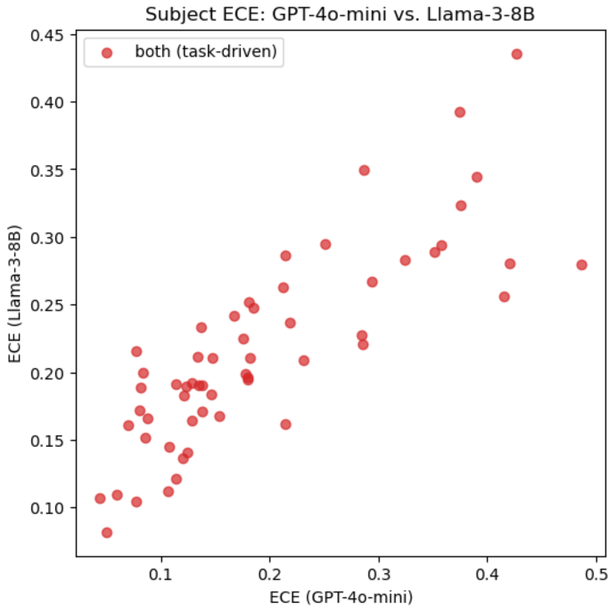

Figure 4: Subject-level ECE for GPT-4o-mini vs. Llama-3-8B-Instruct. Colors indicate agreement: red = both models reject (task-driven), orange = GPT-4o-mini only, blue = Llama only, gray = neither.

of overconfidence varies substantially: GPT-4o-mini shows a 13-fold range (0.037 to 0.487), while Llama-3-8B-Instruct shows a 6-fold range (0.069 to 0.425). Subjects involving ethical reasoning (moral scenarios: 0.487 GPT, 0.279 Llama), formal mathematics (abstract algebra: 0.470 GPT, 0.236 Llama), and legal knowledge (professional law: 0.376 GPT, 0.324 Llama) are most severely affected in both models. This pattern is unlikely to be corrected by a single global temperature parameter, motivating subject-aware or domain-aware recalibration as in Luo et al. [2025].

Cross-model agreement implicates the task, not the model. The Spearman correlation of $\rho = 0.846$ between subject-level ECE rankings, combined with 100% overlap in BH-rejected subjects, provides strong evidence that subject-level miscalibration is task-driven. The same knowledge domains are hardest to calibrate on across both tested models. This has a practical implication: subject-specific miscalibration is likely to persist even with improved models or fine-tuning, since it reflects the inherent structure of the knowledge domains rather than correctable model deficiencies.

Question complexity predicts miscalibration for GPT but not Llama. For GPT-4o-mini, the positive coefficient on entropy ($\hat{\beta} = 0.016$, $p < 0.001$) indicates that when the model is uncertain across options, its overconfidence is *larger*, not smaller. One might expect that distributing probability mass across options would reduce confidence and therefore reduce the gap, but the result suggests the model is miscalibrated even in its expression of uncertainty. Word count is also significant, consistent with longer, more complex questions being harder to reason about. For Llama-3-8B-Instruct, none of these predictors reach significance ($p > 0.05$), though entropy is marginally positive in the same direction ($p = 0.069$). The weaker predictor signal for Llama is consistent with its lower between-subject ICC: with less systematic between-subject structure, question-level features explain less of the remaining variance.

Implications for recalibration. The 7.9% between-subject ICC indicates that a nontrivial fraction of calibration variance is structurally concentrated at the subject level. This has a practical implication: global post-hoc recalibration methods such as temperature scaling, which apply a single correction to all inputs, cannot address this structure. Domain- or subject-specific temperature scaling [Luo et al., 2025] would be needed, at minimum, to correct the within-MMLU heterogeneity documented here.

Limitations. Several limitations warrant caution. First, domain-level variance is estimated at zero, likely because subject-level random effects absorb whatever between-domain structure exists; with only four domain groups, power to detect a separate domain component is very low, so we cannot cleanly partition between-subject variance into domain-level and subject-within-domain components. Second, the mixed model assumes homoskedastic residuals, which is violated when calibration gap variance depends on per-subject accuracy. Third, confidence is computed from logprobs at a single token position; this may not faithfully represent the model's internal uncertainty, especially after RLHF fine-tuning [Kadavath et al., 2022]. Fourth, results are specific to MMLU's four-choice format and may not generalize to open-ended generation settings. Finally, Llama-3-8B-Instruct accuracy (60.6%) is slightly below expected for this model, likely due to 4-bit quantization; this may affect calibration estimates modestly.

Future directions. Subject-specific temperature scaling is a natural next step. Extending the framework to open-ended generation—where confidence elicitation is harder but arguably more consequential—would broaden the audit scope. A fuller cross-model comparison incorporating model scale (e.g., GPT-4o vs. GPT-4o-mini) and fine-tuning method would clarify which aspects of training drive the domain-specific miscalibration patterns identified here.

7 Conclusion

We presented a statistical audit framework for LLM calibration heterogeneity and applied it to MMLU using GPT-4o-mini and Llama-3-8B-Instruct. Both models show universal overconfidence (57/57 subjects rejected at FDR $\alpha = 0.05$) with 13-fold and 6-fold ranges in gap magnitude respectively. Subject-level ECE rankings are strongly correlated across models ($\rho = 0.846$), indicating that miscalibration is task-driven rather than model-specific. GPT-4o-mini shows significant positive predictors of miscalibration (word count, entropy), while Llama shows no significant predictors, consistent with its lower between-subject ICC (2.0% vs. 7.9%). These findings suggest that aggregate ECE substantially understates the heterogeneity of miscalibration across knowledge domains, that subject-aware recalibration is warranted for both models, and that the problematic subjects are unlikely to improve with model-level fixes alone.

Statement of Work

Data collection, statistical analysis, coding, and writing were performed by Jennifer Zhang. John Wright contributed the hierarchical model exposition and additional figure revisions.

References

Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities.* MIT Press, 2023.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1):289–300, 1995.

Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, 2022.

Bradley Efron. *Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction*. Cambridge University Press, 2012.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *Proceedings of the International Conference on Learning Representations*, 2021.

Zhengbao Jiang, Jun Araki, Haibo Ding, and Graham Neubig. How can we know when language models know? On the calibration of language models for question answering. *Transactions of the Association for Computational Linguistics*, 9:962–977, 2021.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Beier Luo, Shuoyuan Wang, Yixuan Li, and Hongxin Wei. Your pre-trained LLM is secretly an unsupervised confidence calibrator. *arXiv preprint arXiv:2505.16690*, 2025.

Preetum Nakkiran, Arwen Bradley, Adam Goliński, Eugene Ndiaye, Michael Kirchhof, and Sinead Williamson. Trained on tokens, calibrated on concepts: The emergence of semantic calibration in LLMs. *arXiv preprint arXiv:2511.04869*, 2025.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022.

Stephen W. Raudenbush and Anthony S. Bryk. *Hierarchical Linear Models: Applications and Data Analysis Methods*. Sage Publications, 2nd edition, 2002.

Hexiang Tan, Wanli Yang, Junwei Zhang, Xin Chen, Rui Tang, Du Su, Jingang Wang, Yuanzhuo Wang, Fei Sun, and Xueqi Cheng. BaseCal: Unsupervised confidence calibration via base model signals. *arXiv preprint arXiv:2601.03042*, 2026.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *Proceedings of the Twelfth International Conference on Learning Representations*, 2024.

Duygu Nur Yaldiz, Evangelia Spiliopoulou, Zheng Qi, Siddharth Varia, Srikanth Doss, and Nikolaos Pappas. Balancing classification and calibration performance in decision-making LLMs via calibration aware reinforcement learning. *arXiv preprint arXiv:2601.13284*, 2026.

Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.